

4 CV of the researcher

Hadrien Lorenzo

PhD

Born on 26/02/1990, in Nice (France), civil partnership, two children

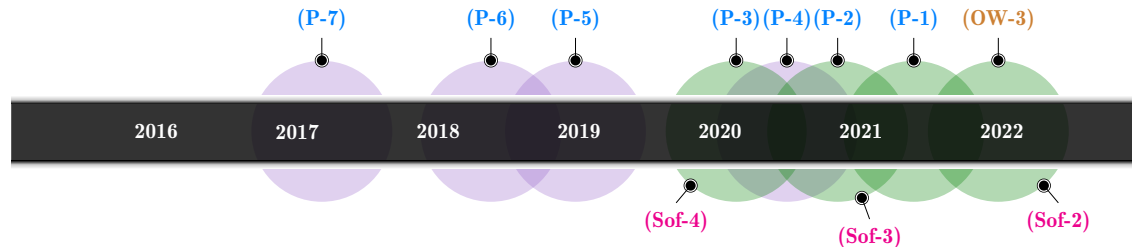
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In this document, (E-★) corresponds to experiences, (P-★) to publications in peer-reviewed journals, (OW-★) to ongoing works and (Sof-★) to software.

Overview of produced research

In the following chart, the circles correspond to approximate dates of peer-reviewed scientific and associated software productions. They can be divided in: **data analysis**, **statistical modelling**.



Professional Experience (E-)

- (E-6) Starts on 01/09/2022: Assistant lecturer at ENSC, the French national graduate school of cognitics. 192 hours of lecturing. Objectives of research are associated with ongoing works (i.e. statistics, probability, regularization and high dimension), see (OW-1) and (OW-2). New collaborations will be made.
- (E-5) 01/09/2020 - 31/08/2022: Post-doctoral researcher at Inria, Astral Team. Project name: *Development of statistical methodologies for high-dimensional multi-block data*. Funding provided by Sartorius, an international company involved in industrial processes regarding to pharmaceutical industry and laboratories. Close collaboration with **Olivier Cloarec**, principal research scientist of Sartorius. The following themes were addressed:
- Supervision of Clément Weinreich on the topic of variable selection in SIR, a non linear semi-supervised approach. Proceeding + under writing paper (OW-2) + package (Sof-1)
 - Statistical definition of the missing value imputation approach **Koh-Lanta** (OW-3) + package (Sof-2). Performed extensive simulations. This follows (E-2).
 - Variable selection and regularization in PLS: **ddsPLS** (P-2) + package (Sof-3). This approach is added to SIMCA, developed by Sartorius (an industry-leading company in procedure monitoring). For SIMCA, their objective is to offer solutions including randomness to users. This follows (E-2).

- Overview of the SIR approach through modern aspects such as high dimension, regularization or variable selection. New collaboration with **Stéphane Girard** (research director, Inria Grenoble, team Statify) the returning supervisor proposed for this action.
- Development of an approach to deal with outliers in SIR: **bootSIR** + package **(Sof-4)**. It uses bootstrap tool. Novelty: use in bag error to classify observations, notion of *learning dynamics*. It can be generalized to any supervised approach **(P-3)**.

(E-4) 01/05/2020 - 31/08/2020: Paternal leave to be with my first child.

(E-3) 01/12/2019 - 30/04/2020: Post-doctoral researcher in SISTM team to pursue different undone projects. Particularly the development of the approach ddsPLS and Koh-Lanta. Those two approaches have been deeply modified during this period and will allow to consolidate those novelties.

(E-2) 01/10/2016 - 30/11/2019: Phd student, under co-direction of **Jérôme Saracco** (Professor, ASTRAL team, Inria) and **Rodolphe Thiébaut** (Professor & Clinical professor, head of the team SISTM, INSERM & Inria), Université de Bordeaux, France. Different topics associated with supervised variable selection in high dimension:

- Analysis of Ebola vaccine (rVSV) research data-set: how to predict the antibody response knowing RNA expression (through RNAseq data: $\approx 20\,000$ variables per sample at days 0, 1, 3 and 7) with 30% of missing values: supervised problem, high dimension, missing data. Application of sPLS for supervised variable selection and Nipals imputation **(P-7)**. Thinking:
 - Sparse PLS with little number of tuning parameters contrary to actual approaches.
 - Impute missing values in supervised context. First time this subject was addressed.
- Prediction of meat tenderness through proteins levels, only 10 cows: high dimension **(P-6)**.
- Analysis of thromboembolic database: 70% of missing values **(P-5)**.

(E-1) 01/10/2015 - 31/09/2016: Research engineer at INSERM. Objectives were: - to familiarize with applied research context in a vaccine oriented research team. - to prospect for a thesis subject with Rodolphe Thiébaut and Jérôme Saracco.

Education

- 27/11/2019: **Phd in Biostatistics**, Université de Bordeaux
Supervised multiblock analysis in high dimension.
Advisors: Jérôme Saracco, Rodolphe Thiébaut.
Rapporteurs: François Husson, Anne-Laure Boulesteix
Committee members: Anne Ruiz-Gazen, Marie Chavent, Arthur Tenenhaus
- 10/09/2015: Double degree:
 - **Msc. Research**, Université Paris-Saclay, *Automation and Signal and Image Processing*.
 - **Msc. Engineering**, CentraleSupélec, *Mathematics and Applied mathematics courses*.

Publications in peer-review journals **(P-)**

In this section, * means same contribution from the authors: **(P-6)** and **(P-7)**. **Acronyms**: IF for Impact Factor, # for “number of citations”.

- (P-1) GIRARD, S., **Lorenzo, H.**, SARACCO, J. “Advanced topics in Sliced Inverse Regression.”, Journal of Multivariate Analysis, 2022, IF=1.009, #=3.
Open-access: <https://hal.inria.fr/hal-03367798/document>
- (P-2) **Lorenzo, H.**, CLOAREC, O., RODOLPHE, T., SARACCO, J. “Data-Driven Sparse Partial Least Squares.”, Statistical Analysis and Data Mining, 2022, IF=1.051, #=0.
Open-access: <https://hal.inria.fr/hal-03368956/document>
- (P-3) **Lorenzo, H.**, SARACCO, J. “Computational Outlier Detection Methods in Sliced Inverse Regression.”, Advances in Contemporary Statistics and Econometrics, *Festschrift in Honor of Christine Thomas-Agnan*, Springer, p. 101-122, 2021, #=1.
Open-access: <https://hal.inria.fr/hal-03369250/document>
- (P-4) ELLIES-OURY, M.-P. ,GRUFFAT, D., LISTRAT, A., **Lorenzo, H.**, CHAVENT, M., SARACCO, J. “The relationships between Sensory and Nutritional Quality are not consistent from one muscle to another in the same bovine carcass.”, Negative Results, 2021.
Open-access: <https://hadrienlorenzo.netlify.app/pdf/negatRes.pdf>
- (P-5) **Lorenzo, H.**, R. MISBAH, J. ODEBER, P.-E. MORANGE, J. SARACCO, D.-A. TRÉGOÛET AND R. THIÉBAUT. 2019, “High-dimensional multi-block analysis of factors associated with thrombin generation potential”, *2019 IEEE 32nd International Symposium on Computer-Based Medical Systems (CBMS)*, IEEE, p. 453–458, #=4.
Open-access: https://hadrienlorenzo.netlify.app/pdf/lorenzo_cbms_2019.pdf
- (P-6) ELLIES-OURY, M.-P.*, **Lorenzo, H.*** DENOYELLE, C., SARACCO, J. AND PICARD, B.. 2019, “An original methodology for the selection of biomarkers of tenderness in five different muscles”, *Foods*, vol. 8, n°6, p. 206, IF=4.305, #=4.
Open-access: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6617152/>
- (P-7) RECHTIEN, A.*, RICHERT*, L. **Lorenzo, H.**, ..., THIÉBAUT, R., ALTFLED, M. AND ADDO, M.. 2017, “Systems Vaccinology Identifies an Early Innate Immune Signature as a Correlate of Antibody Responses to the Ebola Vaccine rVSV-ZEBOV”, *Cell reports*, vol. 20, n°9, p. 2251–2261., IF=9.995, #=89.
Open-access: <https://www.cell.com/action/showPdf?pii=S2211-1247%2817%2931115-4>

Ongoing works (OW-)

- (OW-1) GIRARD, S., **Lorenzo, H.** “Sparse SIR”
- (OW-2) **Lorenzo, H.**, SARACCO, J., WEINREICH, C. “Soft and hard thresholding for variable selection in SIR”
- (OW-3) CLOAREC, O., **Lorenzo, H.**, SARACCO, J. “Koh-Lanta, missing data imputation in supervised context”, Submitted to *Statistical Analysis and Data Mining*

Software productions (Sof-)

- (Sof-1) Package **SIRthresholded** with an available [vignette](#) hosted on a production and on a development platforms:
- (a) on CRAN, the production platform, via <https://cran.r-project.org>, an easy to install solution.

(b) on GitHub, the under development version: [Clement-W/SIR-thresholded](#).

Both come with a full documentation on how to use it, denoted as vignette.

(Sof-2) Package **Koh-Lanta** on GitHub: [hlorenzo/Koh-Lanta](#): call of **ddsPLS** for the supervised part.

(Sof-3) Package **ddsPLS**:

- New R version (**C++** deep optimization code) on GitHub: [hlorenzo/ddsPLS2](#).
- Implementation of multiple imputation Koh-Lanta algorithm.
- R version stable via [CRAN](#) and the associated [vignette](#).
- Python version in development on GitHub: [hlorenzo/py_ddspls](#).

(Sof-4) Package **outlierSIR** on GitHub: [hlorenzo/outlierSIR](#)

(Sof-5) Personal website (R blogdown) via [hadrienlorenzo.netlify.com](#).

Open access dataset

- 29/08/2017: Next generation sequencing RNASeq data set associated with (P-7) from UKE Phase I rVSV ZEBOV vaccine clinical trial, against Ebola, from full blood samples on days 0, 1, 3 and 7 after vaccination. Sensible data so anonymisation was performed. <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE97590>

Funding received so far

- 01/10/2016 - 31/09/2019: Excellence funding for science and medicine researches, a PhD grant for three years. Only three grants this year. “[Thèses Inria-Inserm Santé numérique](#)”, 61 200€ gross amount over three years.
- 01/09/2009 - 31/08/2015: Scholarship on excellence criteria for *Very Good* grade (About 5% of the total secondary school student population), 16 000€ net amount over 8 years.

Collaborations

- [Corporate Research Advanced Data Analytics](#), Sartorius, Bordeaux: [Olivier Cloarec](#) (OW-3), (P-2).
- [Statify team](#), Inria Grenoble Rhône-Alpes, Grenoble, France: [Stéphane Girard](#) (OW-1), (P-1).
- [Astral team](#), Inria Bordeaux Sud-Ouest, Bordeaux: [Jérôme Saracco](#) (OW-2)-(P-6).
- [Biomarkers team](#), INRA, Bordeaux: [Marie-Pierre Ellies](#) (P-4), (P-6).
- [Vintage team](#), INSERM U1219, Bordeaux: [Misbah Razzaq](#), [Felipe Leal Valentim](#) and [David-Alexandre Trégouët](#) (P-5)
- [SISTM team](#), INSERM U1219, Inria BSO, Bordeaux: [Rodolphe Thiébaut](#), [Laura Richert](#), [Solenne Delahaye](#), [Henri Bonnabau](#) and [Boris Hejblum](#) (P-2)-(P-7).
- [University Medical Center Hamburg-Eppendorf](#), 1st Department of Medicine, 20246, Hambourg, Allemagne: [Anne Rechtien](#) (P-7).

Organized Seminars

- Inria, BootSir, 2022, Bordeaux. Organizing board, 10 participants.

- Sanofi Pasteur, a summary of my works, 21st of January 2020, Lyon, France, 15 participants.
- Corporate Research Advanced Data Analytics (Sartorius), How Koh-Lanta works, November 2021, Bordeaux, 12 participants.
- Sanofi Pasteur, a summary of my works, 21st of January 2020, Lyon, France, 15 participants.

Supervising and mentoring activities

- Clément Weinreich, **M1 student**, 3 months project, 2022, development of the **SIR-thresholded** approach, participation to the JDS 2022 and collaboration to **(OW-2)**.
- Arthur Saracco, **L3 student**, 3 months project, 2022, App. development.
- Alexandre Conanec, **PhD student**, steering committee of thesis, title “*Modelling of the optimisation of the management of beef production qualities and performances*”, co-supervision by M.P. Ellies-Oury and J. Sarracco.
- Arthur Saracco, **L3 student**, 1 months internship, June to July 2021. “*Creation of Quizzes to evaluate notions of probability and statistics.*” Co-Supervision with J. Saracco.
- Valentin Kochegarov, **L2 student**, 3 months internship, May to July 2019. “*Etude des méthodes de l’analyse de l’expression différentielle des données de séquençage d’ARN à haut débit.*” Co-Supervision with R. Thiébaud.

Teaching

- **DU BDSI (Big data and statistics for the engineer), ENSC (Bordeaux).** *Courses for adult students.* Class of 15 students. Lecture and tutorials: 2022 (PLS, missing data, 2.5 full days); 2021 (PLS, multiblock analysis, 8 hours); 2019 (PLS, supervised analysis, 11 hours).
- **B.Sc. Engineer, ENSEIRB-MATMECA (Bordeaux), Probabilities and Statistics.** Classes of 30 students: 2022 (22 hours), 2021 (46 hours).
- **M.Sc engineer Biostatistics, ENSAI (Rennes), Genetic project.** Class of 30 students. 2019 (21 hours).
- **M.Sc Biostatistics, ISPED (Bordeaux), Mathematics.** Class of 30 students. 2019: Lecture and tutorial (6 hours). 2018: Lecture and tutorial (26 hours).
- **M.Sc. Health and Science mixed-master (Bordeaux), the objective was to quickly form medical students to “Mathematics” (Analysis, Algebra, Probabilities, Statistics and Coding).** Classes of 30 students, 2019: Lecture and tutorial (24 hours).
- **ISPED Summer School (Bordeaux).** Class of 30 students. 2018: Lecture and tutorial (6 hours).

Involvement in learned societies

- Member of the SFDS (*Société Française De Statistiques* $\approx$ French Statistical Society) since 2016.

5 Capacity of the Participating Organisations

5.1 Overview of Participating Organisations

Organisation role	PIC	Legal Entity Short Name	Academic organisation (Y/N)	Country	Name of Supervisor
Beneficiary & secondment venue for the first 2 months of the outgoing phase	999547074	INRIA	Y	France	Stéphane Girard
Associated partner for outgoing phase	957524443	EPM	Y	Canada	Julie Carreau

5.2 Capacity of the Participating Organisations

Associated partner for outgoing phase	
CORPORATION DE L ECOLE POLYTECHNIQUE DE MONTREAL ASSOCIATION + EPM + Canada	
General description: Founded in 1873, Polytechnique Montréal (PM) is one of Canada's largest engineering teaching and research institutions and has educated over 52 500 engineers. It ranks first in Québec in terms of the scope of its research activities. Its annual research budget stands at \$110 million and its 60 research units allow its teams of professors. The institution accounts for the production of nearly one-quarter of university engineering research in Québec.	
Role and profile of supervisor	Julie Carreau (OS) is Assistant Professor in the MAGI department (2021-) and is also affiliated with IVADO, Institute for Data VALorization. She has a mixed background in mathematics and machine learning. She has spent 14 years in France, initially as a postdoctoral fellow and then with a tenure position at <i>Institut de Recherche pour le Développement</i> (IRD). As a postdoctoral fellow, she worked with the RS on a topic linked to the SUMEXT proposal. The OS will provide both research training and personal tuition on various topics associated with the SUMEXT project. She will also give the PF the opportunity to increase its network through its numerous collaborations in Montreal.
Key research facilities, Infrastructure and Equipment	PM will provide means of calculation thanks to CalculQuébec but also courses in order to handle different protocols/languages (MPI, OpenMP, CUDA, OpenACC). Assistance on arrival, finding accommodation and integration of the family and spouse. The PF will have a desk (if available), a whiteboard, an internet connection and access to scientific papers from all major publishers.
Previous and current involvement in EU funded research and training programmes, actions, projects	• 31st Annual Conference of the EAIE, Helsinki, Finland, 09/2019 (European Association for International Education). • PM is Member of the CLUSTER network: a pan-European group of University of Science and Technology universities. • PM is Member of the T.I.M.E Association: Top International Managers in Engineering, with mainly European universities. • Julie Carreau was personally involved in European funded research projects directly associated with the SUMEXT project (see <i>Part A</i>, Section 2 - <i>Participants</i>).

Beneficiary & secondment venue for the first 2 months of the outgoing phase	
INSTITUT NATIONAL DE RECHERCHE EN INFORMATIQUE ET AUTOMATIQUE + INRIA + France	
General description: Established in 1967, Inria is a French public research institute dedicated to the entire spectrum of computational sciences, producing outstanding research with strong economic and societal impact (more than 200 companies have stemmed from Inria). The institute's strength is its ability to develop a culture of scientific innovation and stimulate creativity in digital research (more than 70 ERC grants since 2007, #1 European institute in the area of digital technology). Inria was awarded the European Union's HRS4R label in 2019, recognising its commitment to provide the best environment for its researchers. The Inria Grenoble center was created in 1992 and currently hosts a workforce more than 600 people, including 230 PhD students and Post-Docs. STATIFY is a joint team of Inria and Laboratoire Jean Kuntzmann, a joint research unit of the CNRS, Institut National Polytechnique de Grenoble (G-INP) and UGA. The team is composed of five permanent members, thirteen PhD students and one post-doc. The team is engaged in several international collaborations and has developed strong ties with the industry such as EDF, Valeo or Criteo. The team also transferred its knowledge through the spin-off start-up Pixyl.	
Role and profile of supervisor	Stéphane Girard (RS) is Research Director (1st grade, DR1) in the STATIFY team and, from 2018, part-time associate researcher at Ecole Polytechnique. He is/was associate editor of internationally recognized scientific journals. He is a specialist of extreme-value theory (EVT) and high-dimensional statistics. Stéphane Girard published more than 120 research papers in top-ranked journals including <i>Annals of Statistics</i>, <i>Journal of Machine Learning Research</i>, <i>Bernoulli</i>, <i>JRSS-B</i>, with 5000+ citations and h-index 40 from google scholar. The RS will provide the academic secondment to the PF. During this two month period, the PF will be trained on EVT. This will be a training-through-research period working on the first objective of the SUMEXT project. The RS will also integrate the PF to the scientific meetings of the team and favouring new collaborations with members of this team and other teams of Inria and UGA.
Key research facilities, Infrastructure and Equipment	The PF will have a desk, a whiteboard, an internet connection and access to scientific papers from all major publishers. The PF will have access to the means of calculations available at Inria such as JeanZay and Plafrim.
Previous and current involvement in EU funded research and training programmes, actions, projects	• SIMERG2E (Associated teams, Statistical Inference for the Management of Extreme Risks and Global Epidemiology, 2015-2021): Association with LERSTAD (Université Gaston Berger, Sénégal). The team involves EQUIPPE (Université de Lille) laboratory members and associated members of Modal (Inria Lille) as well as an epidemiologist from IRD at Dakar. Two research themes: Spatial extremes with application to management of extreme risks; Classification for global epidemiology. • Risk@UGA (IDEX, 2018-2021): Cross-disciplinary research and scientific innovation in the field of disaster and risk management, in areas vulnerable due to interdependencies between human, natural or technological hazards. Specific areas: Grenoble, Beirut, Port-au-Prince, Peru and Nepal. The project also aims at proposing a risk institute within UGA. It federates a hundred researchers. • Data@UGA (IDEX, 2017-2020): Data change science & society. Three research themes: spatial and environmental sciences, biology, health sciences; data-driven research in Social Sciences and Humanities; data security and privacy of data. In collaboration with Inria Team Necs and Gipsalab and with IPAG and Univ. Paris Sud Orsay. 8 keuros awarded from the IRS call: research project on extreme and Bayesian statistics. • HUMAVIPS (FP7-ICT, European STREP project, 2010-2013, 2.62M€): Humanoids with auditory and visual abilities in populated spaces. Objective: develop humanoid robots + audio-visual perception systems + social skills + handling multi-party conversations and interactions with people in realtime. Involved partners: Perception and STATIFY (Inria, coord.), Czech Technical University (CTU), Aldebaran Robotics ALD France, IDIAP research institute in Switzerland and Bielefeld University (BIU) in Germany. • Vahinés (ANR, 2008-2012): This three-year project is called "Visualisation et analyse d'images hyperspectrales multidimensionnelles en Astrophysique" (VAHINES). It aims at developing physical as well as mathematical models, algorithms, and software able to deal efficiently with hyperspectral multi-angle data & other kind of large hyperspectral dataset (astronomical or experimental). It involves the Observatoire de la Côte d'Azur (Nice), and universities (Strasbourg I and Grenoble I). • MedUp (ANR, 2008-2011, 834 524 euros, 36 months): "Forecast and projection in climate scenario of Mediterranean intense events: Uncertainties and Propagation on environment" and deals with the quantification and identification of sources of uncertainties associated with the forecast and climate projection for Mediterranean high-impact weather events. involves Météo-France and several universities (Paris VI, Grenoble I and Toulouse III).

6 Additional ethics information

The Section **6. Non-EU Countries** detailed the question:

Will some of the activities be carried out in non-EU countries?

The answer is **Yes** since 24 months of the action will be carried out in Montréal (Canada).

7 Additional information on security screening

There are no additional information on security screening associated with the SUMEXT project.

8 Letter of commitment from associated partners

The letter of commitment for the outgoing phase is available on the next page.

Montreal, September 02, 2022

To Whom it May Concern,

I, the undersigned, Pr. Yuvin Adnarain CHINNIAH, in my quality of Director of the Mathematical and Industrial Engineering Department at Polytechnique Montréal, commit to set up all necessary provisions (as detailed below) to participate as associated partner in the proposal SUMEXT submitted to the call HE-MSCA-PF-2022, should the proposal be funded.

On behalf of Polytechnique Montréal, I confirm that we will participate and contribute to the research, innovation and training activities as planned in this project. In particular, Polytechnique Montréal will be involved in hosting Dr. Hadrien Lorenzo as an experienced researcher for 24 months during the period of 2023 – 2025. Dr. Hadrien Lorenzo proposes to work on probabilistic modeling of multiple output extreme regression, estimators and associated asymptotic and consistency results. Dr. Hadrien Lorenzo plans to apply these developments to the context of extreme precipitation and storm surge.

This work will be done under the supervision of Prof. Julie Carreau, assistant professor in the department of Mathematical and Industrial Engineering of Polytechnique Montreal, as part of her research program on the statistical modeling of environmental and climatic processes involving the Ouranos consortium, Environment and Climate Change Canada and IVADO. Prof. Jonathan Jalbert, associate professor in the department of Mathematical and Industrial Engineering of Polytechnique Montreal, will also be involved in Dr. Hadrien Lorenzo's project. Dr. Hadrien Lorenzo will have the opportunity to attend graduate mathematical courses to improve his knowledge in theoretical probabilities at either Polytechnique Montréal, Université de Montréal or HEC Montréal. I also agree to provide Dr. Hadrien Lorenzo an office (if space is available) and free access to computer resources at Polytechnique Montréal for the duration of his stay.

I hereby declare that I am entitled to commit into this process the entity I represent.

Best regards,



Pr. Yuvin Adnarain CHINNIAH
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